

# Geometry of Particle Methods

Jordan, Kinderlehrer, Otto 1998: reinterpret Langevin dynamics as a *minimizing movement* scheme:

$$\mu_{k+1} := \arg \inf_{\mu} \text{KL}(\mu, \pi) + \frac{1}{2\eta} W_2^2(\mu, \mu_k) \quad \pi, \mu_k \in \mathcal{P}(\mathbb{R}^d)$$

First insights into rich theory of *Wasserstein Geometry*, a “Riemannian” calculus on  $\mathcal{P}(\mathbb{R}^d)$ !

# Geometry of Particle Methods

## Definition (First Variation)

Fix  $\mathcal{F} : \mathcal{P}(\mathbb{R}^d) \rightarrow \mathbb{R}$ . The first variation at  $\mu \in \mathcal{P}(\mathbb{R}^d)$  is (when it exists) the function  $\delta\mathcal{F} : \mathbb{R}^d \rightarrow \mathbb{R}$  which satisfies,

$$\frac{d}{d\epsilon}\mathcal{F}(\mu + \epsilon\xi) = \int \delta\mathcal{F}[\mu](x) \xi(dx)$$

for all  $\mu$  with finite second moment and all signed measures  $\xi$  with  $\int \xi(dx) = 0$ .